

Roll No.

TCE-305

B. TECH. (CIVIL ENGG.) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

ENGINEERING MECHANICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

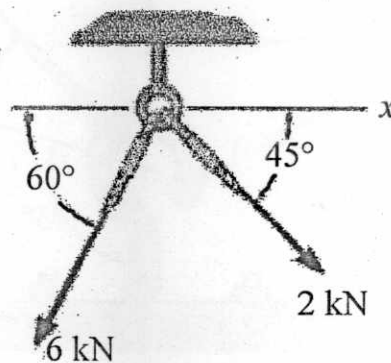
(ii) Each sub-question carries 10 marks.

1. (a) Explain the principle of superposition and principle of transmissibility of forces.

(CO1)

OR

- (b) Determine the magnitude of the resultant force acting on the screw eye and its direction measured clockwise from the x-axis : (CO1)

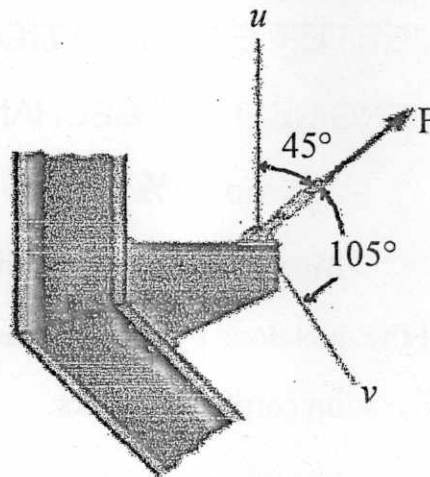


P. T. O.

2. (a) Define moment and couple of a force. Write down characteristics of a force. (CO1)

OR

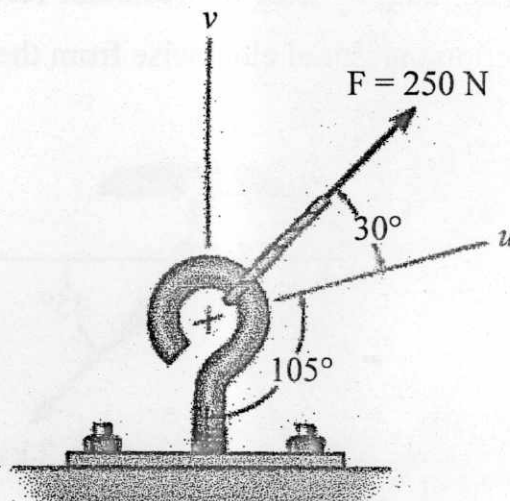
- (b) If force F is to have a component along the u -axis of $F_u = 6$ kN, determine the magnitude of F and the magnitude of its component F_v along the v -axis. (CO1)



3. (a) State Newton's three laws. (CO1)

OR

- (b) Resolve F into components along the u and v -axes and determine the magnitudes of these components : (CO1)



(3)

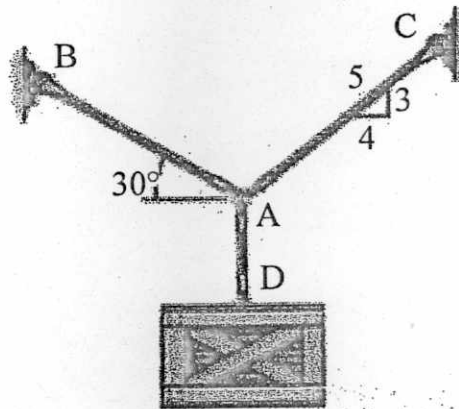
4. (a) Explain Lami's theorem.

(CO2)

OR

- (b) The crate has a weight of 550 N. Determine the force in each supporting cable.

(CO2)



5. (a) Define coplanar and concurrent forces.

(CO2)

OR

- (b) In each case, write two equations of equilibrium. Do not solve.

(CO2)

